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Evidence-Based IP Network-Type Classification

Shriraj Bhargave
[shrirajchaitanya.ai25@bmsce.ac.in]

Shreyas Mulekar
[shreyasmulekar.ai25@bmsce.ac.in]

Professor Manjula M
[manjulam.csi@bmsce.ac.in]

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Problem Description

Static IP databases misclassify users; we need classifications that can be justified.

- Fraud engines and content platforms label IPs as residential, data-center, commercial VPN or Tor; coarse static tables often misjudge legitimate users.
- Shared-address architectures (RFC 6269) mean an IP does not map cleanly to one user or one type of network.
- Goal: classify each timestamped IP observation as Known Tor exit, Known hosted VPN/proxy endpoint, or Unknown.
- Every label must ship with an audit log: which evidence, from which source, as of which snapshot.
- “Unknown” is a first-class answer whenever evidence is inconclusive. No device identity, OS inference, or claim that an Unknown IP is a direct connection.

Background

What is needed to understand the problem and our approach.

- IP addresses are shared and reassigned (NAT, CGNAT, hosting pools), so a label is only valid for a point in time [1].
- ASN/BGP metadata maps a prefix to its originating AS at a given moment; routing data changes continuously [2].
- Tor publishes exit relay lists; relays join and leave, so membership is time-bound [3].
- Hosting/data-center ASNs host many things besides VPNs: hosting alone is not proof of VPN/proxy use.
- Evaluation uses an organiser-curated labelled laboratory dataset with known endpoints.

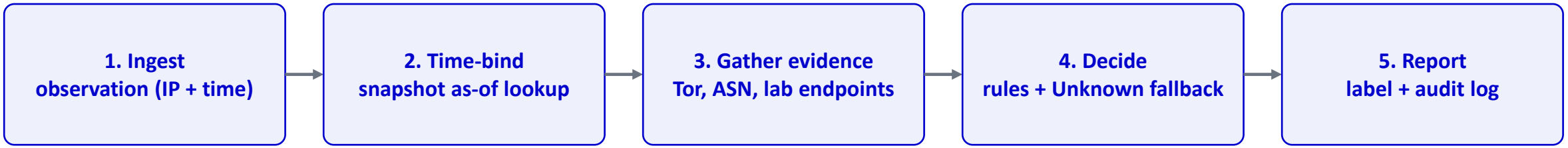
Literature & RFC Review

Only the most relevant standards and data sources.

Source	What it gives us	How we use it
RFC 6269 [1]	Issues with IP address sharing	Justifies time-bound, evidence-only labels; no per-user claims
RFC 4271 [2]	BGP-4: origin AS of a prefix	Timestamped prefix-to-ASN lookups
Tor exit lists [3]	Published exit relay addresses	Primary evidence for the Known Tor exit class
RFC 6811 [4]	RPKI origin validation	Optional signal on the trustworthiness of the ASN origin

Proposed Solution

A rule-based, evidence-first pipeline with an explicit Unknown outcome.



- Tor exit: address is in a published exit list valid at the observation time.
- Hosted VPN/proxy: address matches a known lab endpoint, with ASN/hosting records as corroborating evidence.
- Unknown: no qualifying evidence, conflicting evidence, or stale snapshot. Hosting-only evidence is not enough.
- Each decision stores source, snapshot timestamp and matched rule as provenance.

Optimization Proposed

Before vs. after compared with a static IP-database approach.

Aspect	Static database (before)	Evidence-based pipeline (after)
Time handling	One table, no validity window	As-of lookups against dated snapshots
Uncertain cases	Forced into a class	Returns Unknown with the reason
Explainability	Label only	Audit log: evidence, source, timestamp
Hosting IPs	Often tagged VPN/proxy	Hosting alone stays Unknown; tested as negatives
Quality reporting	Rarely published	Per-class precision, recall and coverage

Solution Architecture & Design

Visual: component view of the classification process.

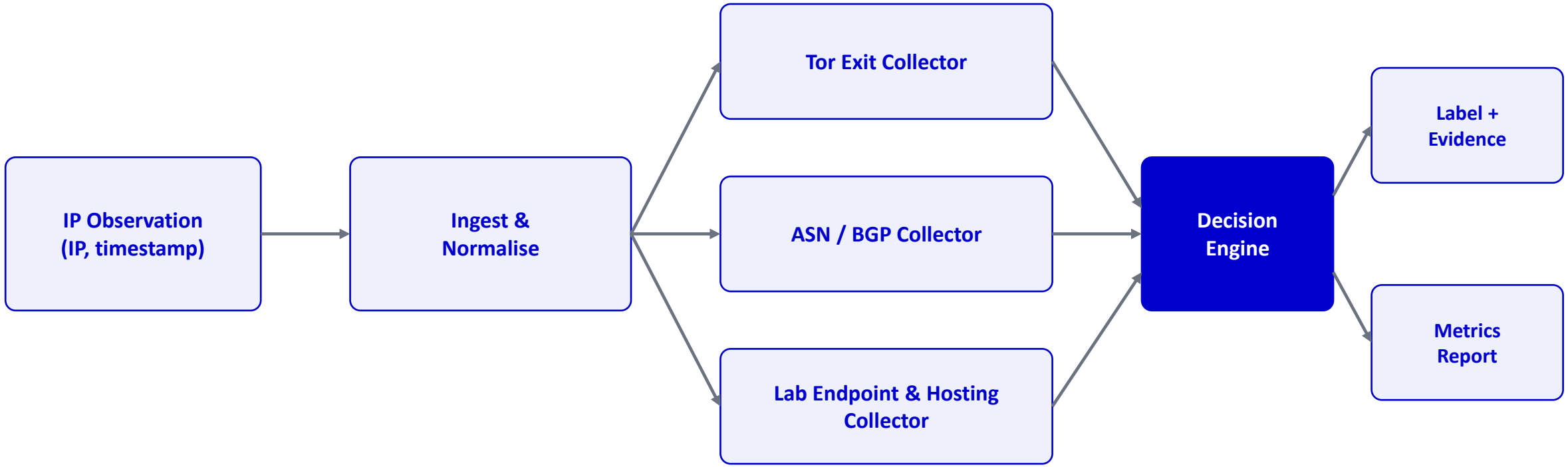


Figure: System architecture (UML component view)

Data Flow Diagram

Overview of the data flow, from observation to audited result.

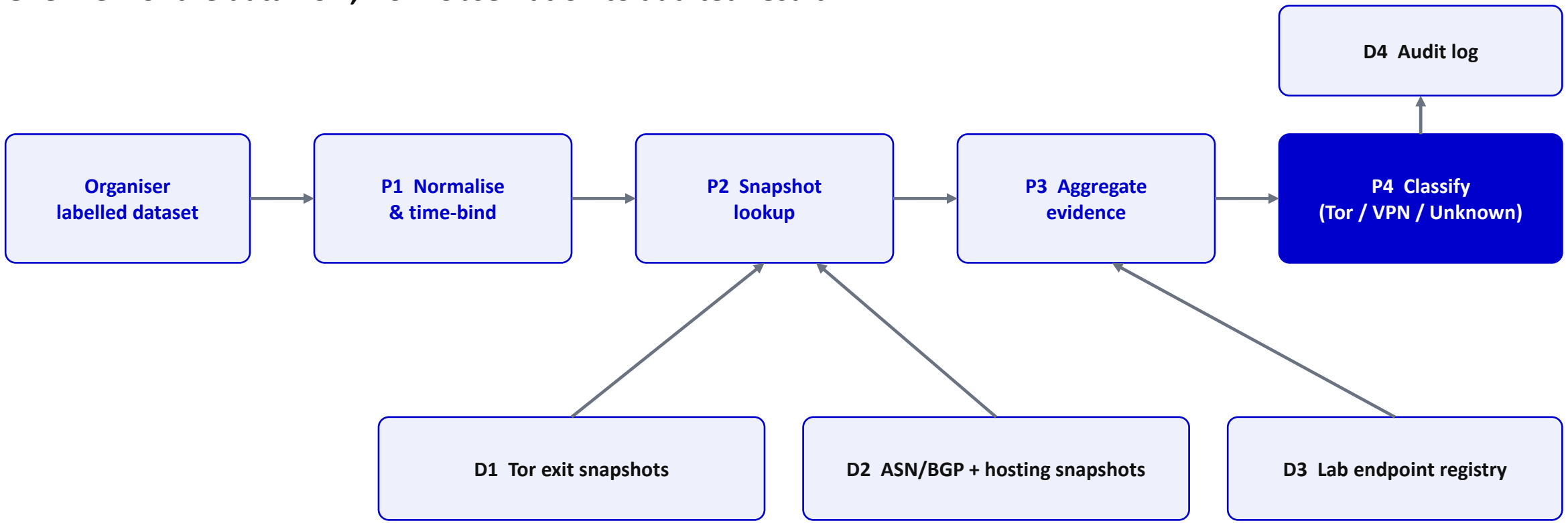


Figure: Data flow diagram (DFD). Lookups use data valid at the observation timestamp.

Deployment Timeline

Milestones and methodology: iterative, evidence tested against the labelled set at each step.

Phase	Milestone	Output
1. Data	Collect dated Tor, ASN/BGP and hosting snapshots; load lab endpoints and dataset	Versioned snapshot store
2. Classifier	Implement time-bound rules, evidence logging and Unknown handling	Working classifier + audit log
3. Evaluation	Per-class precision, recall, coverage; hosting-IP negative tests	Metrics report
4. Delivery	Final PDF in repo, mentor demo with pseudocode snippet	Technical report + demo

Bibliography

[1] M. Ford, M. Boucadair, A. Durand, P. Levis, P. Roberts. “Issues with IP Address Sharing.” RFC 6269, IETF, June 2011.

[2] Y. Rekhter, T. Li, S. Hares (Eds.). “A Border Gateway Protocol 4 (BGP-4).” RFC 4271, IETF, January 2006.

[3] The Tor Project. “Tor bulk exit list and relay data.” https://check.torproject.org/torbulkexitlist

[4] P. Mohapatra et al. “BGP Prefix Origin Validation.” RFC 6811, IETF, January 2013.

Thank You for Your Attention!